

POI Graphs under Spatial Noise and Temporal Drift

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Abstract—Next Point-of-Interest (POI) recommendation is a central task in location-based social networks, aiming to predict a user’s future destinations from historical mobility trajectories. Recent advances in graph neural networks and sequential modeling have significantly improved recommendation accuracy by incorporating spatial transitions, contextual information, and high-order behavioral dependencies. However, real-world mobility data are often affected by GPS inaccuracies, sparse observations, and temporal distribution shifts caused by seasonal behavior changes, venue popularity dynamics, and evolving user preferences. Existing evaluations are commonly conducted on heavily preprocessed datasets with occurrence-based filtering and static train-test assumptions, making the robustness of modern POI recommendation methods under realistic conditions insufficiently understood.

This project investigates the robustness of next-POI recommendation methods under spatial perturbation and temporal drift. Rather than proposing a new state-of-the-art architecture, we develop a systematic evaluation framework for comparing representative sequential and graph-based recommendation models under varying preprocessing settings and controlled spatial perturbations. Experiments will be conducted on publicly available Foursquare and Gowalla datasets using standard ranking metrics including Recall@k and NDCG@k, together with robustness-oriented measures based on relative performance degradation.

The goal of this work is to provide practical insights into the reliability and deployment readiness of modern POI recommendation systems in noisy and non-stationary environments.

Index Terms—POI recommendation, GNN, spatial noise, robustness evaluation, comparative analysis

I. INTRODUCTION

LOCATION-BASED social networks (LBSNs) such as Foursquare and Gowalla have enabled the large-scale collection of human mobility data through user check-ins and location-sharing services. These data have driven substantial research interest in next Point-of-Interest (POI) recommendation, a task that aims to predict the next location a user is likely to visit based on historical trajectories. Accurate next-POI recommendation has important applications in urban computing, personalized services, tourism, advertising, and intelligent transportation systems.

Early approaches to next-POI recommendation primarily relied on Markov chains and recurrent neural networks to model sequential mobility behavior. More recently, graph-based methods have become dominant due to their ability to capture complex transition patterns and collaborative relations between POIs. Graph neural networks (GNNs), attention mechanisms, and Transformer-based architectures have demonstrated strong performance by modeling spatial proximity, trajectory transitions, and contextual dependencies across user mobility graphs. Recent work has further explored multi-graph fusion and contrastive learning to improve

representation quality under sparse check-in data.

Although recent POI recommendation methods incorporate sophisticated graph structures and self-supervised learning objectives, relatively little attention has been devoted to robustness evaluation under spatial perturbation and temporal drift. Existing studies typically employ random or short-horizon train-test splits and evaluate models on clean benchmark datasets, which may overestimate real-world generalization performance or result in simple difference in training data between models. As a result, it remains unclear which architectural mechanisms contribute most to robustness under noisy and non-stationary conditions.

This work addresses this gap through a systematic comparative study of representative sequential and graph-based next-POI recommendation methods under controlled spatial and temporal distribution shifts. Instead of introducing another complex recommendation architecture, the primary goal of this project is to evaluate how existing methods behave when deployed with varying noise levels.

II. RELATED WORK

Contrastive learning has recently emerged as an effective self-supervised learning strategy for rec-

ommendation systems. The general idea is to generate multiple augmented views of graph-structured data and learn representations by maximizing agreement between positive pairs while separating negative samples.

Several foundational methods introduced contrastive objectives for graph recommendation. SGL [1] applies node and edge dropout to generate augmented graph views, while SimGCL [2] introduces perturbations directly in the embedding space to improve robustness and representation consistency. NCL [3] further incorporates neighborhood-enriched contrastive objectives to capture structural information more effectively.

In the POI recommendation domain, contrastive learning is commonly used to alleviate sparsity and improve multi-view representation learning. CLLP [4] combines spatiotemporal preference modeling with contrastive objectives, while MGMCL [5] aligns representations across multiple mobility graphs through local and global contrastive learning strategies.

Although these approaches improve representation quality and generalization, contrastive augmentations are typically designed for representation learning rather than systematic robustness evaluation. In particular, the effects of controlled spatial perturbation and temporal distribution shift on next-POI recommendation models remain insufficiently explored.

III. DATASETS AND PREPROCESSING

We use three real-world check-in datasets:

- **TSMC2014_NYC**: 227,428 check-ins, 21,000 POIs, 1,083 users (Foursquare, Apr 2012–Feb 2013) [6]
- **TSMC2014_TKY**: 537,703 check-ins, 18,000 POIs, 2,293 users (Foursquare, Apr 2012–Feb 2013) [6]
- **Gowalla (global)**: 6.4M check-ins, 1.28M POIs, 107,092 users with check-ins (Feb 2009–Oct 2010) [7]

A. Exploratory Data Analysis

a) *Gowalla*: We analyzed the full SNAP Gowalla check-in dataset before graph construction. The raw file contains check-ins with user IDs, timestamps, coordinates, and POI IDs over the period February 2009–October 2010.

TABLE I
CLEANED GOWALLA DATASET SUMMARY

Statistic	Value
Raw check-ins	6,442,892
Cleaned check-ins	6,442,262
Users	107,092
POIs	1,280,956
Category coverage by POI	96.8%
Category coverage by check-in	97.3%
POI name coverage by POI	0.5%
POI name coverage by check-in	0.2%

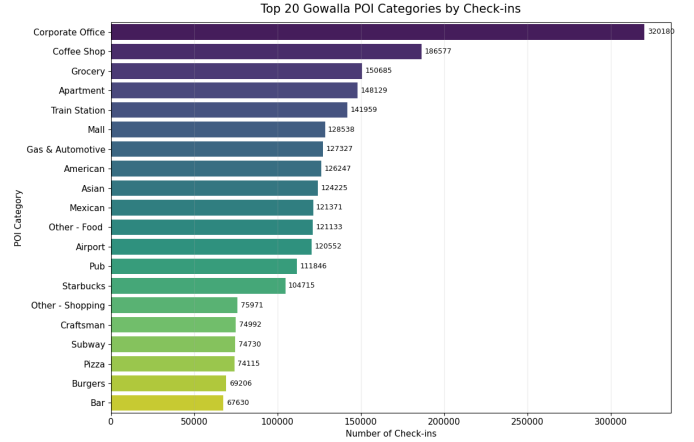


Fig. 1. Top Gowalla POI categories by number of check-ins.

SNAP Gowalla does not include POI names or semantic categories, so we enriched POIs with Figshare Gowalla metadata [8]. Category metadata covers 96.8% of POIs and 97.3% of check-ins. POI names cover only 0.5% of POIs and 0.2% of check-ins; therefore, the analysis focuses mainly on POI categories.

Fig. 1 shows the most frequent POI categories. The leading categories include corporate offices, coffee shops, grocery stores, apartments, train stations, malls, and transportation-related venues. This indicates that Gowalla contains heterogeneous urban mobility patterns rather than one dominant type of location.

Fig. 2 shows row-normalized hourly profiles for the most frequent categories. Each row sums to one, so the figure compares the daily rhythm within each category. The profiles differ substantially across categories, which confirms that time-of-day information is relevant for next-POI modeling.

The POI distribution is highly long-tailed (Fig. 3). In the cleaned dataset, 17,886 POIs cover 25% of

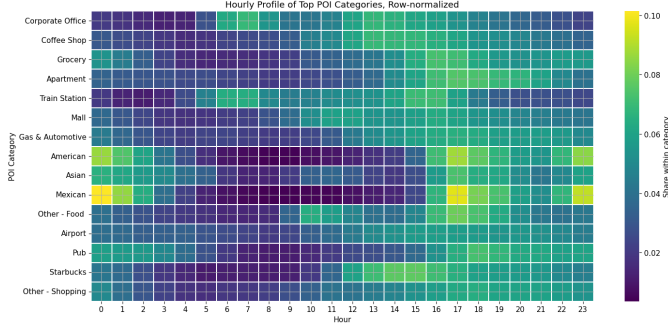


Fig. 2. Hourly profile of top POI categories, row-normalized within each category.

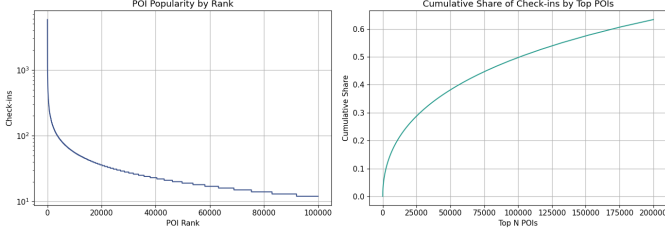


Fig. 3. POI popularity long tail and cumulative check-in coverage.

check-ins, 101,432 cover 50%, 346,882 cover 75%, and 709,806 cover 90%. This confirms strong sparsity at the POI level and motivates filtering, graph augmentation, and careful treatment of infrequent locations.

Additional EDA results show that user activity is also uneven: most users have relatively few check-ins, while a small subset is much more active. Repeat-visit analysis shows that trajectories contain both returns to previously visited POIs and exploration of new locations. We also constructed consecutive POI transitions and measured transition time gaps and distances. These results indicate that transition graph construction should account for temporal windows and spatial distance, rather than treating all consecutive check-ins as equally meaningful mobility transitions.

Fig. 4 summarizes temporal drift. Monthly active users and active POIs change substantially over time, while the overlap of top POIs across adjacent months is not fully stable. This supports chronological train/validation/test splitting instead of random splitting, since future mobility behavior may differ from past behavior.

Overall, the EDA shows three properties relevant to our study: high category coverage, strong long-tail sparsity, and temporal drift. These properties

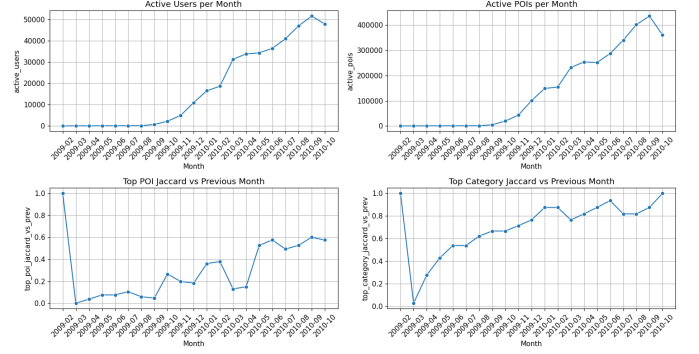


Fig. 4. Temporal drift in monthly Gowalla activity and top-entity stability.

TABLE II
BASIC DATASET STATISTICS

Metric	New York City	Tokyo
Total Check-ins	227,428	573,598
Unique Users	1,083	2,293
Unique Venues	38,333	61,858
Unique Categories	251	247
Avg Check-ins per User	210.0	250.2
Median Check-ins per User	153	173
Min Check-ins per User	100	100
Max Check-ins per User	2,697	2,991

motivate robustness evaluation under spatial noise and temporal distribution shift. Additional EDA tables, preprocessing checks, and visualizations are provided in the supplementary notebook [9].

b) Foursquare: We also considered Foursquare [6] which is a common LBSN dataset. This dataset contains check-ins in NYC (New York City) and TKY (Tokyo) collected for about 10 month (from 12 April 2012 to 16 February 2013). Each check-in is associated with its time stamp, its GPS coordinates and its semantic meaning (represented by fine-grained venue-categories).

The general statistics (Table II) show that Tokyo contains roughly twice as much data as NYC, with both user count and total check-ins approximately doubling. Foursquare appears to have applied some filtering based on minimum check-ins per user, though the threshold appears to be relatively low, so this should not pose a concern for analysis.

A more detailed examination of user activity distributions (Fig. 5) reveals that NYC and Tokyo exhibit rather similar patterns, although Tokyo users appear slightly more active. Both datasets contain anomalous users with over 2,500 check-ins.

Category-based behavior differs significantly be-

tween the two cities. The top categories (Fig. 6), shown both by total check-ins and unique users, reveal that rankings not only differ across regions but are also substantially influenced by highly active users. For instance, the 'Office' category ranks high in NYC but is absent from Tokyo's top list. Similarly, bars appear as a top priority for New York users, yet rank at best fifth in Tokyo (after Japanese restaurants). The difference between ranking methods also reveals variation in user activity patterns. Train stations in Tokyo rank first by a large margin (~150K check-ins) when considering total check-ins, but when measured by unique users, they surpass Japanese restaurants by only a few hundred. This suggests that different venue types exhibit distinct user engagement patterns — it is reasonable to assume that a user checking into a train station or subway would do so twice per workday, unlike entertainment venues.

Examining hourly, daily, and monthly activity patterns (Fig. 7), the data shows clear dependence on time of day and month. User activity decreases at night and exhibits pronounced peaks during commuting hours and lunch breaks. The data collection spans from April to February, showing peaks from April to July, followed by a decline in August and September, with no data for October. A second period of increased activity occurs from November to January.

We also examined whether activity patterns vary by category. The row-normalized heatmaps (Fig. 8) show that subway and train stations in both NYC and Tokyo peak in the morning, with NYC users arriving at work shortly after. Bars in New York are popular from after work until dawn, while Tokyo residents frequent ramen shops and Japanese restaurants both for lunch and after-work dinner. Comparing hourly activity between weekdays and weekends (Fig. 13) shows that check-in spikes generally occur on weekdays at similar times in both cities.

Inspecting temporal patterns more closely using a 7-day rolling window (Fig. 12) reveals similar daily check-in patterns for NYC and Tokyo. User activity spikes in the early months of 2012 (April–July), declines from July to October, and then exhibits another spike with a more stable trend line. This may indicate rising app popularity due to advertising campaigns or other external factors; in any case, visible time-based trends are present.

The relationship between categories and the number of unique venues (Fig. 10) appears similar between NYC and Tokyo, with points concentrated in an almost linear relationship and some anomalous high-outlier points present in both datasets. Comparing category choice diversity among users (Fig. 11), most users fall below 3 bits in both datasets, with medians of 2.9 for NYC and 2.38 for Tokyo.

A minor additional observation: HDBSCAN clustering of POIs (Fig. 9) reveals numerous small clusters, often elongated and likely following street patterns, with both cities exhibiting concentrated urban centers.

Overall, the EDA shows that in both NYC and Tokyo user activity follows power-law distributions. The data exhibits commuting peaks, dramatic weekend shifts and seasonal effects. However between them category preferences diverge sharply, with NYC also winning by the level of user activity and diverse venues and categories. Full code and results can be found in open source [10].

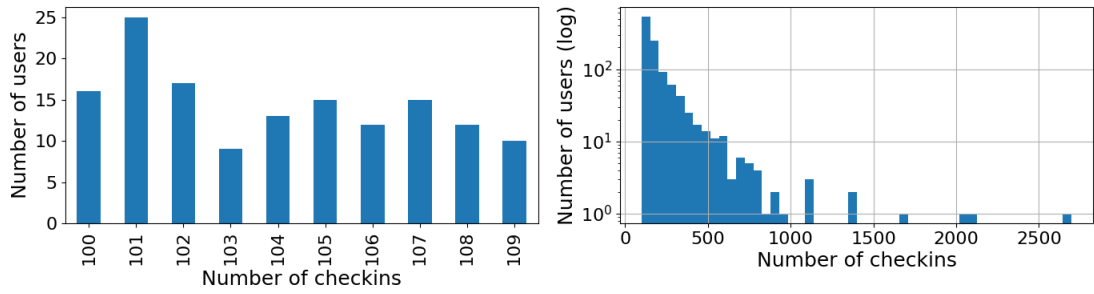
IV. CONCLUSION

REFERENCES

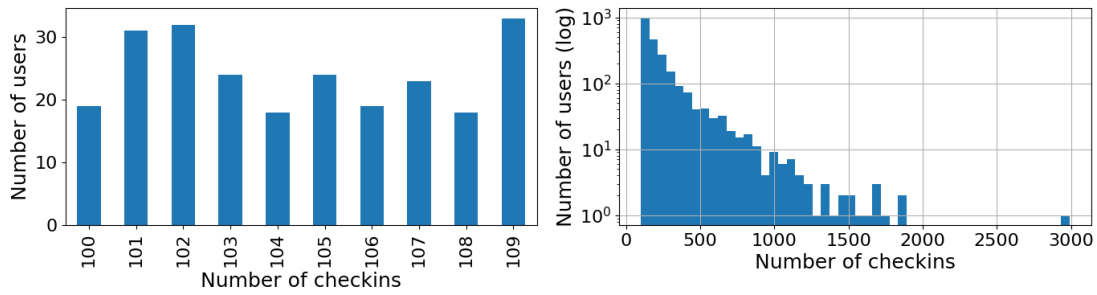
- [1] J. Wu, X. Wang, F. Feng, X. He, L. Chen, J. Lian, and X. Xie, "Self-supervised graph learning for recommendation," in *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2021, pp. 726–735.
- [2] J. Yu, H. Yin, X. Xia, T. Chen, L. Cui, and Q. V. H. Nguyen, "Are graph augmentations necessary? simple graph contrastive learning for recommendation," in *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2022, pp. 1294–1303.
- [3] Z. Lin, C. Tian, Y. Hou, and W. X. Zhao, "Improving graph collaborative filtering with neighborhood-enriched contrastive learning," in *Proceedings of the ACM Web Conference 2022*, 2022, pp. 2320–2329.
- [4] H. Zhou, Z. Jia, H. Zhu, and Z. Zhang, "CLLP: Contrastive learning framework based on latent preferences for next POI recommendation," in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024, pp. 1473–1482.
- [5] Y. Wang and Q. Jiang, "Enhancing next POI recommendation via multi-graph modeling and multi-granularity contrastive learning," *Journal of King Saud University - Computer and Information Sciences*, vol. 37, p. 343, 2025.
- [6] D. Yang, D. Zhang, V. W. Zheng, and Z. Yu, "Modeling user activity preference by leveraging user spatial temporal characteristics in LBSNs," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 45, no. 1, pp. 129–142, 2015, <https://sites.google.com/site/yangdingqi/home/foursquare-dataset>.

- [7] E. Cho, S. A. Myers, and J. Leskovec, “Friendship and mobility: User movement in location-based social networks,” in *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2011, pp. 1082–1090.
- [8] “gowalla_data,” https://figshare.com/articles/dataset/gowalla_data/22126586/1, 2023, figshare dataset, accessed May 22, 2026.
- [9] M. Urazmetova, “Full gowalla exploratory data analysis with figshare POI metadata,” https://colab.research.google.com/drive/1vbVnU8Fj_wbxYSYV9KXu_lGtEu3dQtBD?usp=sharing, 2026, supplementary notebook.
- [10] U. Klyuchnikova, “Full foursquare tkynyc exploratory data analysis,” <https://www.kaggle.com/code/alice6fox/foursquaregraphs-eda>, 2026, supplementary notebook.

APPENDIX



(a) NYC: Check-ins per user



(b) Tokyo: Check-ins per user

Fig. 5. Distribution of user activity levels. Left: Top-10 users by check-in count. Right: Log-scale histogram showing power-law behavior.

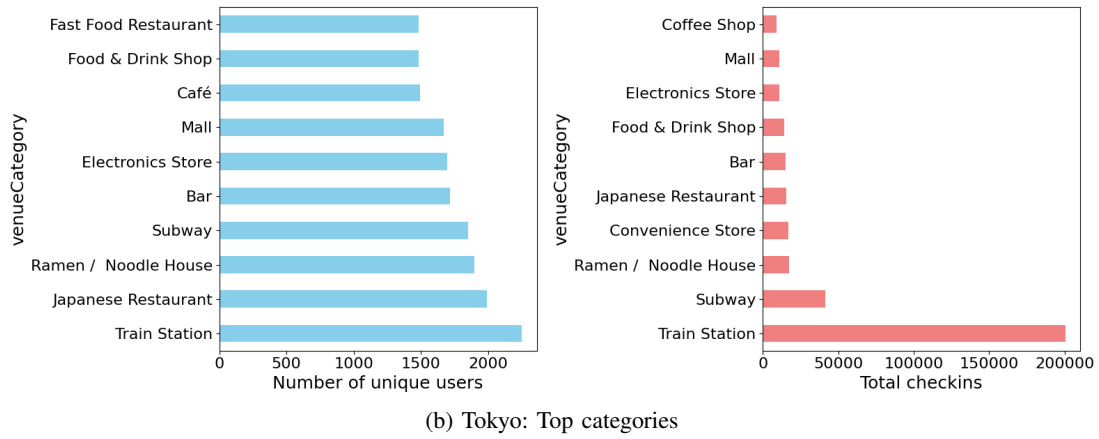
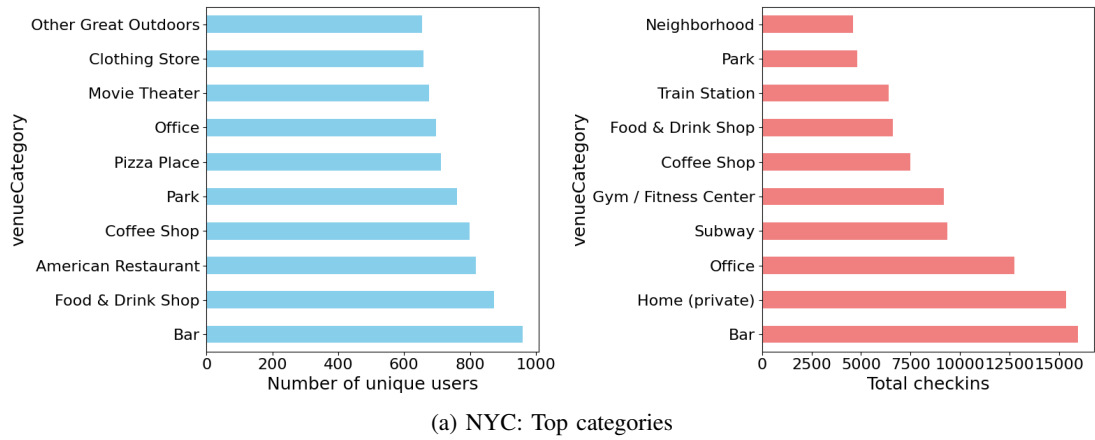


Fig. 6. Top-10 venue categories. Left: By unique users (reach). Right: By total check-ins (frequency).

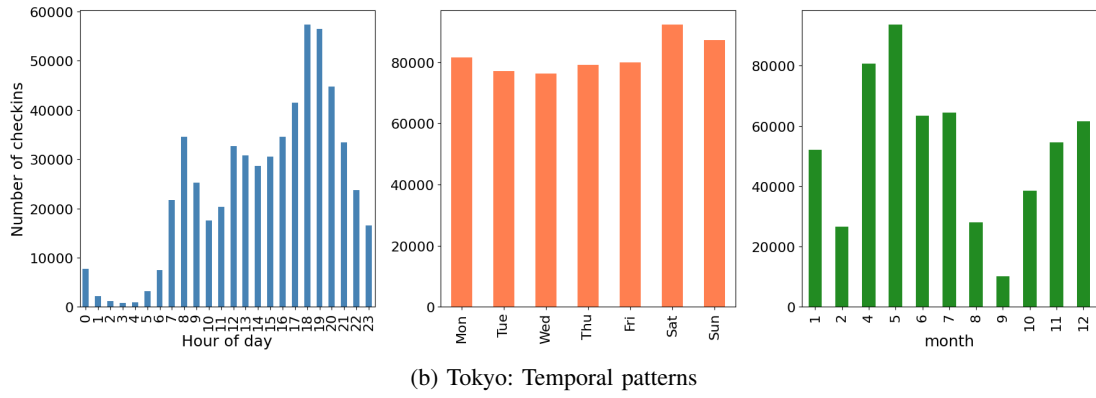
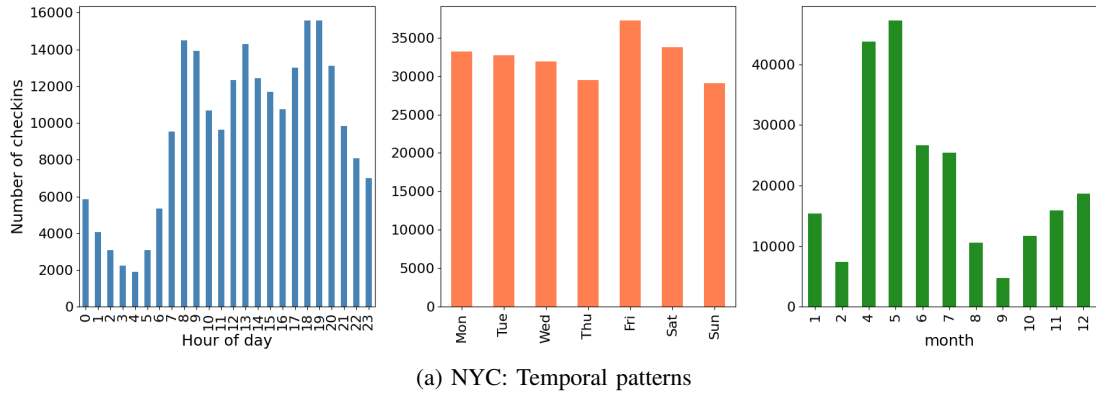
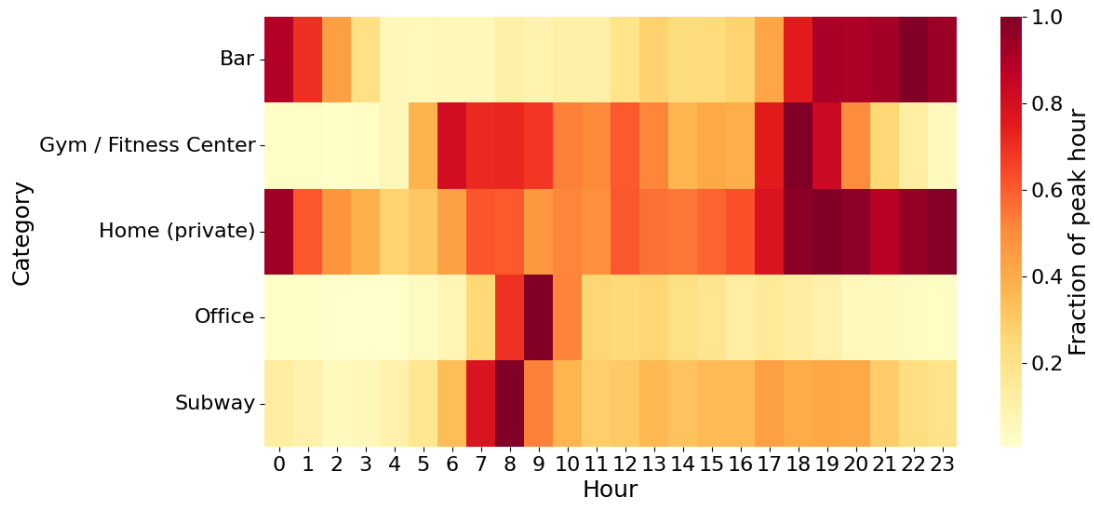
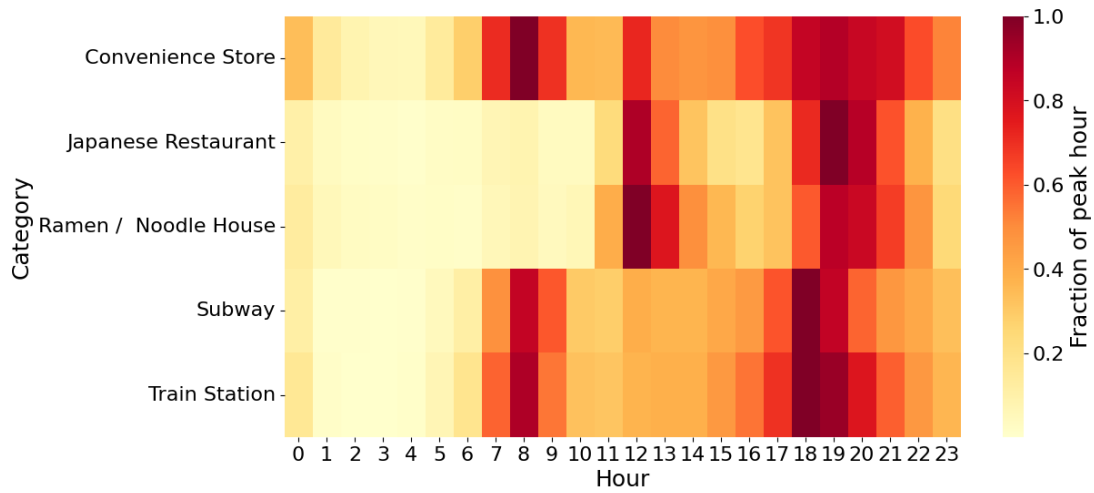


Fig. 7. Temporal patterns: Hourly, daily, and monthly distributions.



(a) NYC: Category-hour heatmap



(b) Tokyo: Category-hour heatmap

Fig. 8. Category activity by hour (row-normalized). Darker color = peak hour for that category.

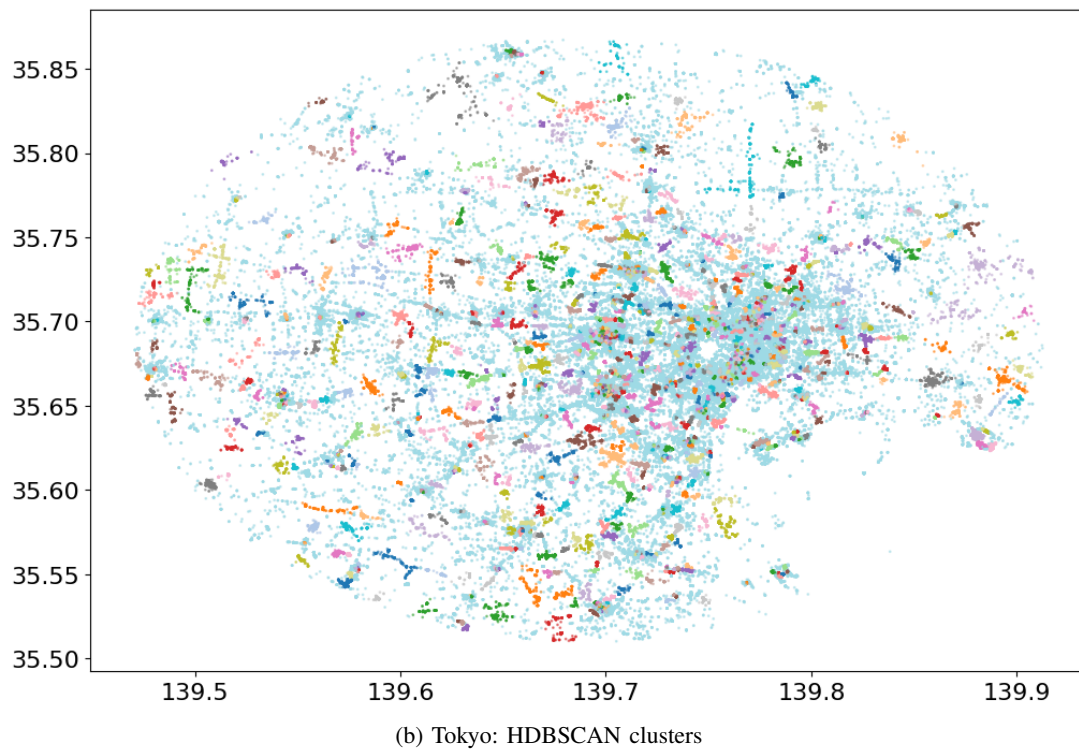
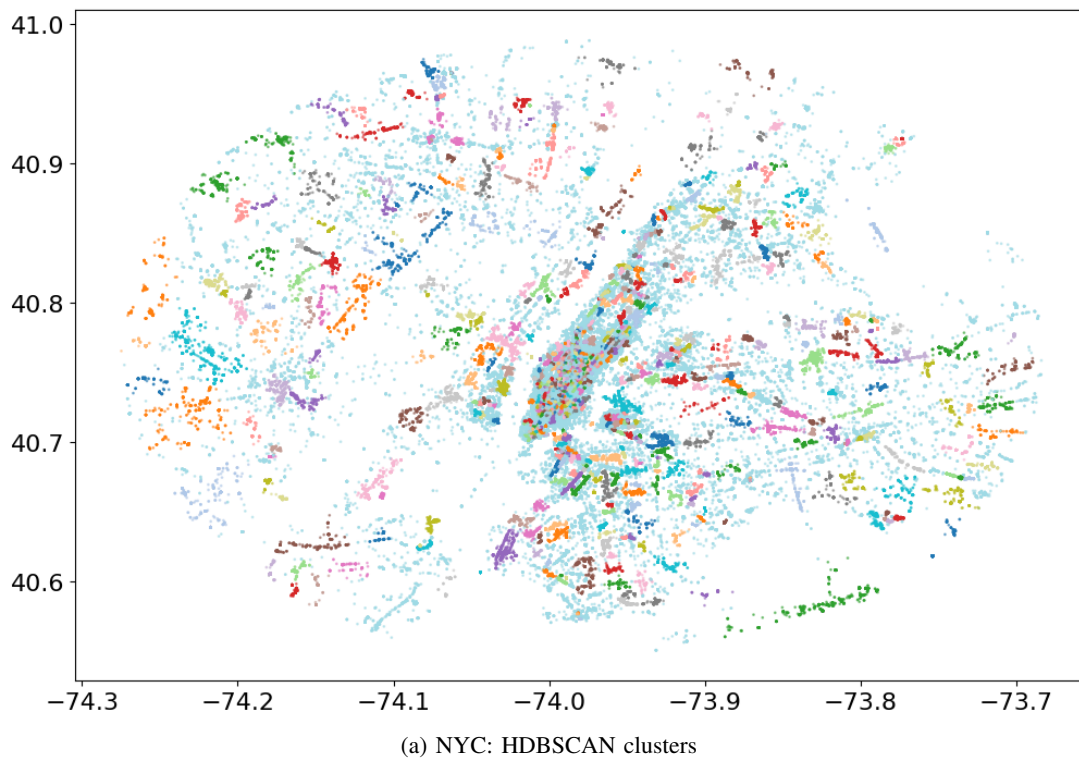
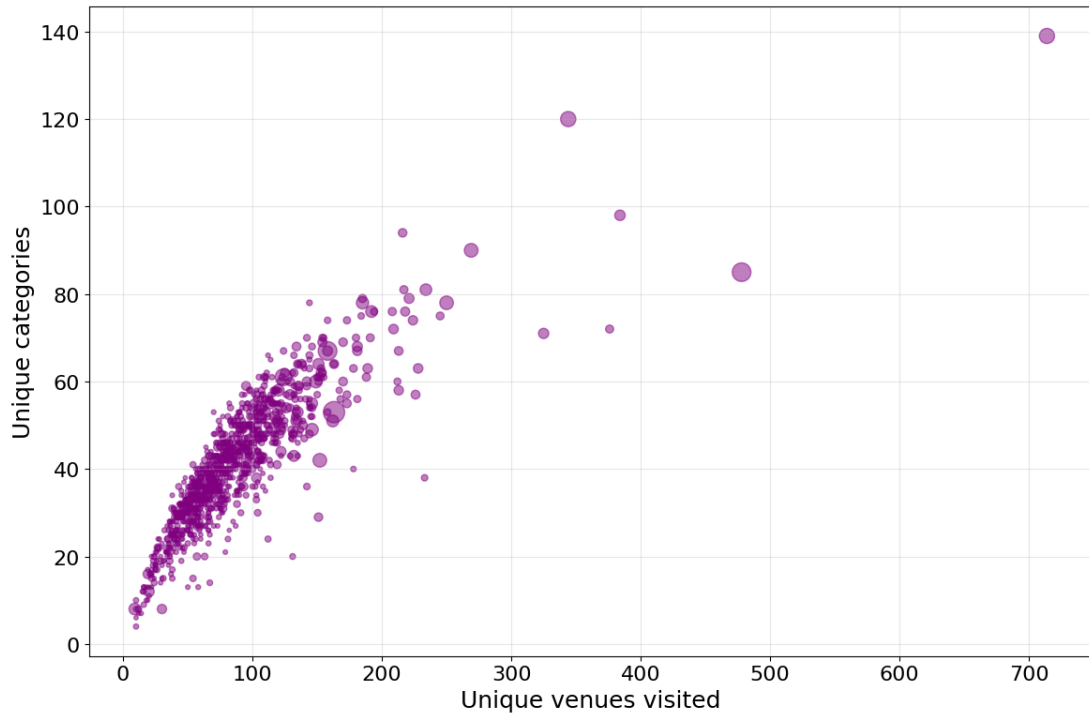
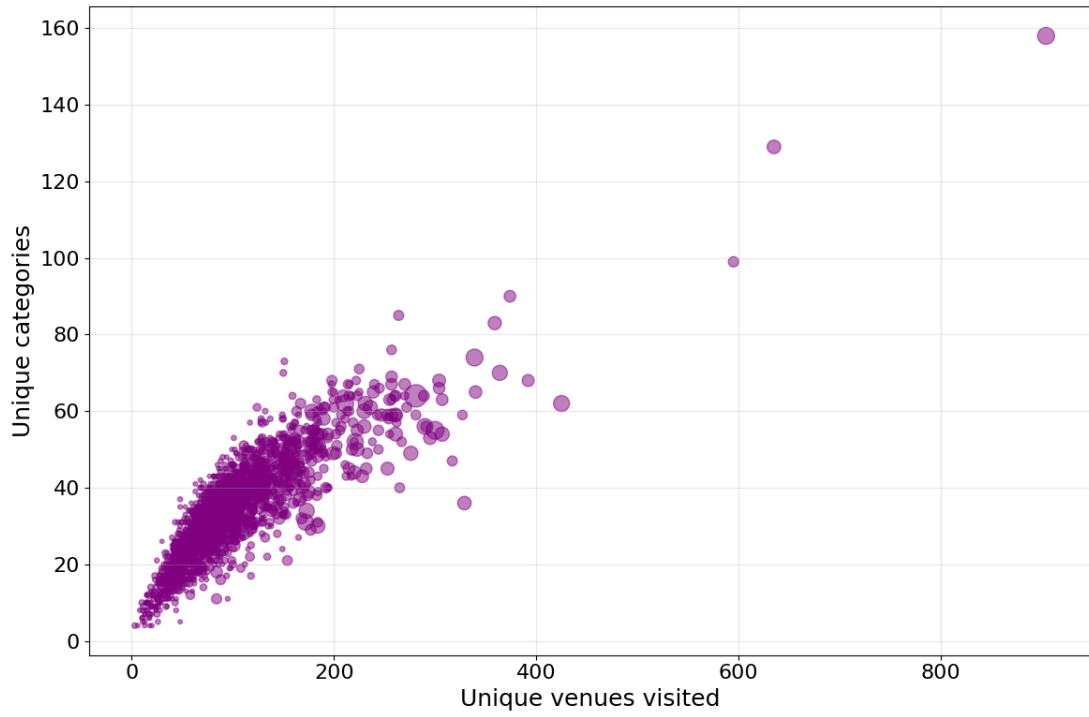


Fig. 9. Geographic clustering of check-ins. Each color represents a distinct cluster; gray points are noise.

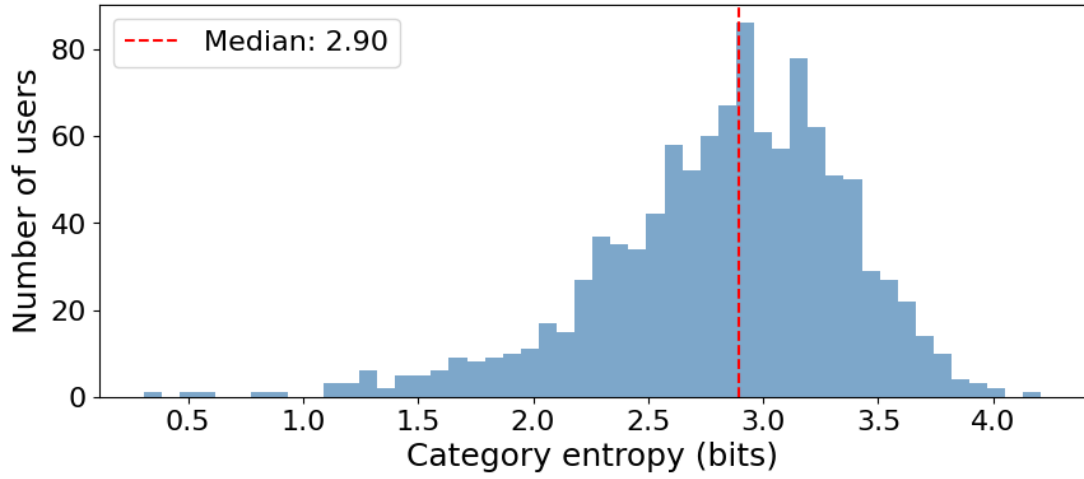


(a) NYC: Venues vs Categories

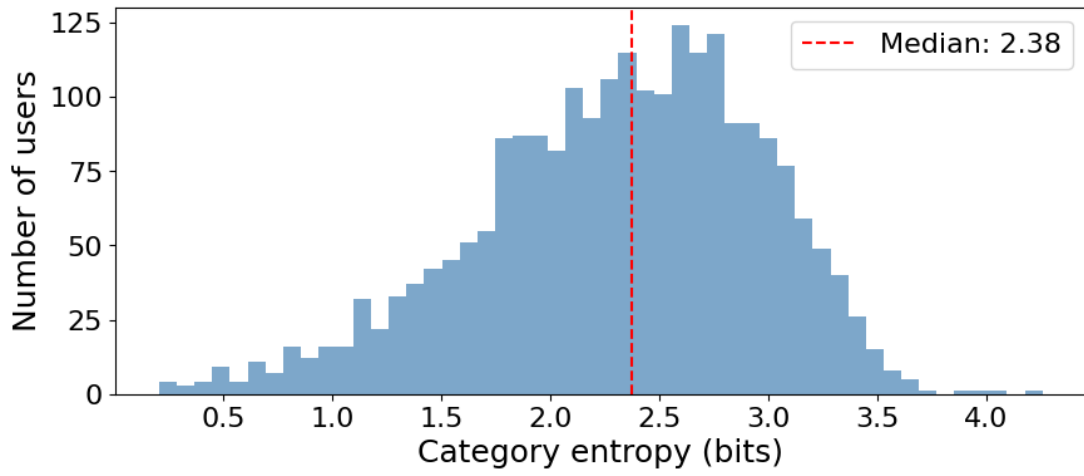


(b) Tokyo: Venues vs Categories

Fig. 10. User diversity: Unique venues visited vs unique categories explored. Point size = total check-ins.

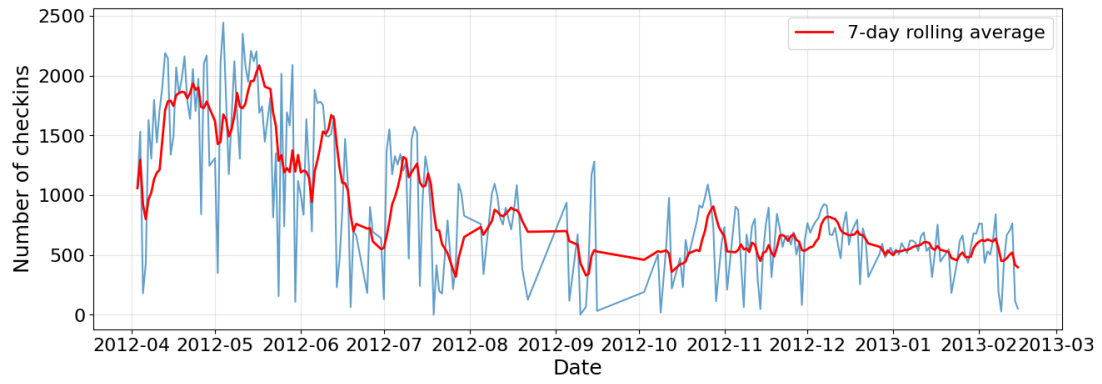


(a) NYC: Category entropy

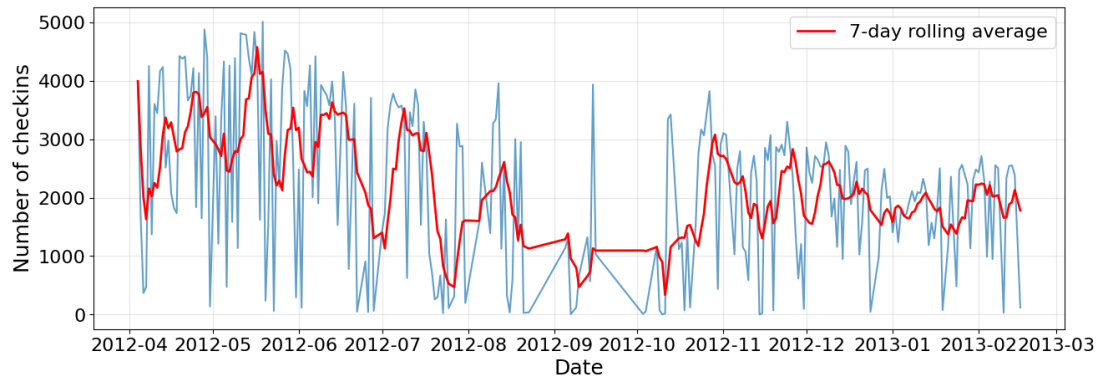


(b) Tokyo: Category entropy

Fig. 11. User category diversity measured by Shannon entropy (bits). Higher entropy = more diverse exploration.

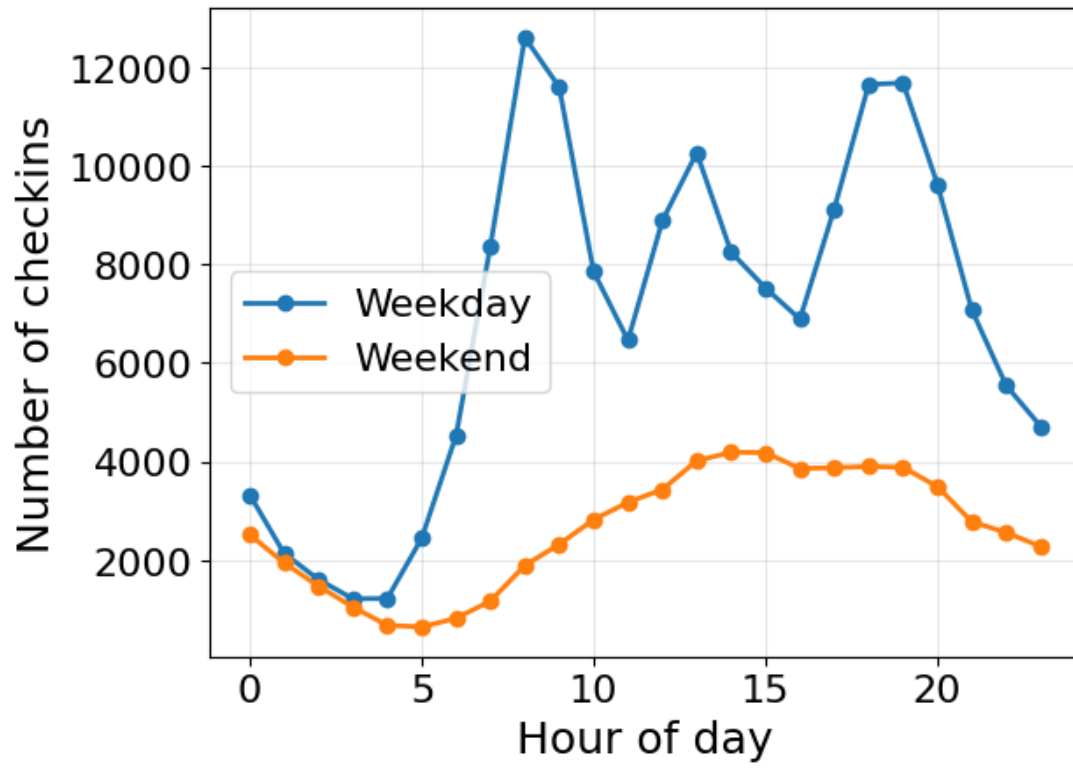


(a) NYC: Daily check-ins

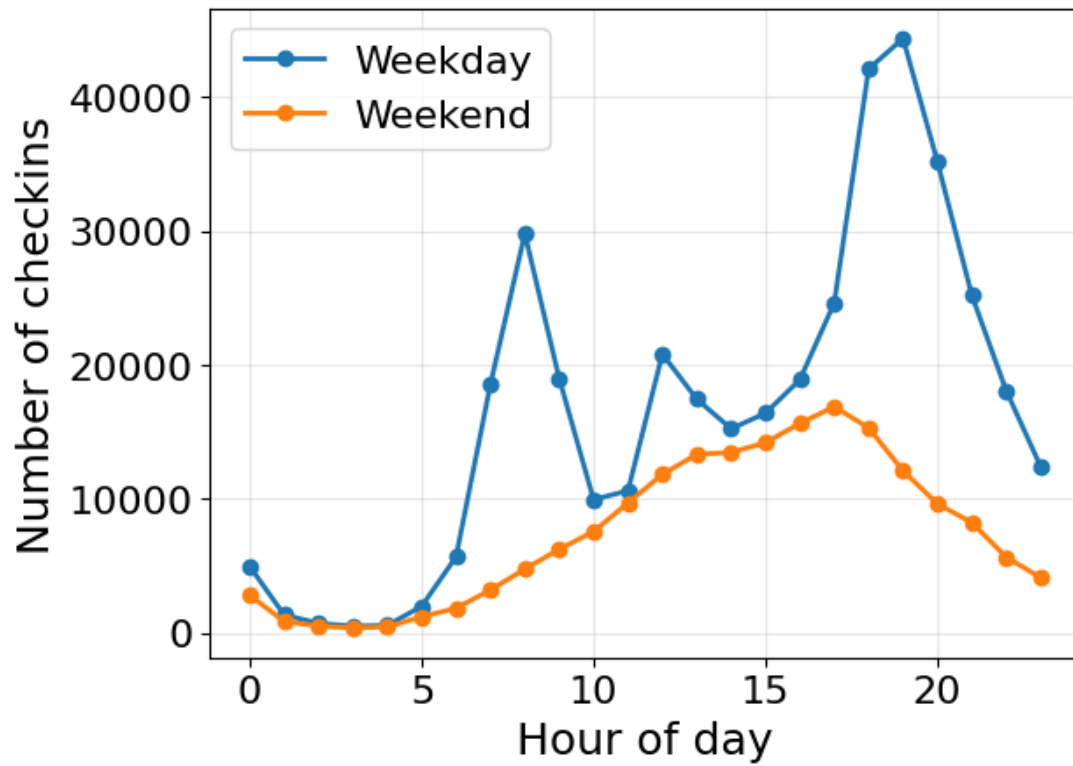


(b) Tokyo: Daily check-ins

Fig. 12. Daily check-ins with 7-day rolling average (red).



(a) NYC: Weekday vs Weekend



(b) Tokyo: Weekday vs Weekend

Fig. 13. Weekday vs weekend hourly patterns.